Problem Set for 19-May and 22-May

Part I: the Kronecker product, tensor products for matrices

Example Take

- $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}, X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}, Y = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix}$. Now

1. The matrix equation $AX = Y$ coincides the linear system

$$\begin{pmatrix} a_{11} & 0 & a_{12} & 0 \\ 0 & a_{11} & 0 & a_{12} \\ a_{21} & 0 & a_{22} & 0 \\ 0 & a_{21} & 0 & a_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

2. The matrix equation $XB = Y$ coincides linear system

$$\begin{pmatrix} b_{11} & b_{21} & 0 & 0 \\ b_{12} & b_{22} & 0 & 0 \\ 0 & 0 & b_{11} & b_{21} \\ 0 & 0 & b_{12} & b_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

3. The matrix equation $AXB = Y$ coincides linear system

$$\begin{pmatrix} a_{11}b_{11} & a_{11}b_{21} & a_{12}b_{11} & a_{12}b_{21} \\ a_{11}b_{12} & a_{11}b_{22} & a_{12}b_{12} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{21} & a_{22}b_{11} & a_{22}b_{21} \\ a_{21}b_{12} & a_{21}b_{22} & a_{22}b_{12} & a_{22}b_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

   where we write

$$\begin{pmatrix} a_{11}b_{11} & a_{11}b_{21} & a_{12}b_{11} & a_{12}b_{21} \\ a_{11}b_{12} & a_{11}b_{22} & a_{12}b_{12} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{21} & a_{22}b_{11} & a_{22}b_{21} \\ a_{21}b_{12} & a_{21}b_{22} & a_{22}b_{12} & a_{22}b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}B^T & a_{12}B^T \\ a_{21}B^T & a_{22}B^T \end{pmatrix} =: A \otimes B^T.$$

Definition (Kronecker product of matrices) For $P \in K^{k\times l}, Q \in K^{m\times n}$, define

$$P \otimes Q = \begin{pmatrix} p_{11}Q & p_{12}Q & \cdots & p_{1l}Q \\ p_{21}Q & p_{22}Q & \cdots & p_{2l}Q \\ \vdots & \vdots & \ddots & \vdots \\ p_{k1}Q & p_{k2}Q & \cdots & p_{kl}Q \end{pmatrix}.$$

Clearly, $P \otimes Q \in K^{km \times ln}$.

Exercise 1 Verify the following statements yourself (**not a homework**):

1. $\otimes$ is an binary operation which is associative;
2. $\otimes$ is bilinear on both sides;
3. $(A \otimes B) \cdot (C \otimes D) = (A \cdot C) \otimes (B \cdot D)$;
4. $(A \otimes B)^T = A^T \otimes B^T$;
5. $\operatorname{rank}(A \otimes B) = \operatorname{rank}(A) \cdot \operatorname{rank}(B)$;

Now we assume that A and B are square matrices:

1. $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$;
2. $\det(A \otimes B) = (\det A)^{\dim A} (\det B)^{\dim B}$;
3. $\operatorname{tr}(A \otimes B) = \operatorname{tr}(A) \cdot \operatorname{tr}(B)$;
4. Let (λ, v) and (μ, u) be eigenpairs of A and B, respectively. Then $A \otimes B$ has $(\lambda\mu, v \otimes u)$ as an eigenpair;

Exercise 2 Let J_A and J_B denote the Jordan form of A and B respectively (all matrices are in $\mathbb{F}^{n \times n}$).

1. Write down the Jordan form of the linear transformation $X \mapsto AXA^T$;
2. Write down the Jordan form of the linear transformation $X \mapsto AXA$;
3. Write down the Jordan form of the linear transformation $X \mapsto AX - XA$;
4. Write down the Jordan form of the linear transformation $X \mapsto AX - XA^T$;
5. Write down the Jordan form of the linear transformation $X \mapsto AXB$.

> **Tip**
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> We know that $A \sim A^T$ over arbitrary field, since $(\lambda I - A)$ and $(\lambda I - A^T)$ are equivalent.

Exercise 3 Show that $\dim \ker[X \mapsto (AX - XA^T)] \geq n$, and explain when the equality holds.

Exercise 4 Show that there exists $\{0, 1\}$-matrices A and B such that $A \otimes B$ and $B \otimes A$ are not similar.

> **Tip**
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> For square matrices A and B, it is clear that $A \otimes B \sim B \otimes A$. Proof of the statement: By linear isomorphism of $(-)^T$, $[X \mapsto AXB^T]$ and $[Y \mapsto BYA^T]$ stand for the same linear map under different basis. Hence $(A \otimes B) \sim (B^T \otimes A^T)$. Since $A \sim A^T$ and $B \sim B^T$, one has $(A \otimes B) \sim (B \otimes A)$.

Exercise (optional) If you know the adjacency matrices $A(G_i)$ for a simple graph G_i, then discover the graph operations $\boxplus$ and $\boxtimes$ such that

1. $A(G_1) \otimes A(G_2) = A(G_1 \boxtimes G_2)$;
2. $A(G_1) \otimes I + I \otimes A(G_2) = A(G_1 \boxplus G_2)$.

> **Tip**
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> Kronecker 积 $\otimes$ 的本质是指标的运算，图论语言可以将这一指标运算可视化.

Part II: the tensor product in general, and the use of $1:1$ correspondence

> **Important**
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> Throughout, $\sum$ is always a finite sum.

Definition (Tensor product of two finite dimensional vector spaces). Let $\{u_i\}_{1\leq i\leq n}$ be a basis of U, and $\{v_j\}_{1\leq j\leq m}$ be a basis of V. Now one can show the $1:1$ correspondence of the following three types of functions.

1. a bilinear map from U, V to $\mathbb{F}$;
2. a map of the type $f: S \to \mathbb{F}$ with property P, where
   - $S := \{M \in \mathbb{F}^{n\times m} \mid \mathrm{rank}(M) = 1\}$, the subspace of rank one matrices;
   - P means that, once there is an equality of linear combination in S (e.g., $\sum c_i s_i = 0$ for $s_i \in S$), one has $\sum c_i f(s_i) = 0$.
3. a linear map $\mathrm{Hom}_{\mathbb{F}}(\mathbb{F}^{n\times m}, \mathbb{F})$.

We identify $U \otimes V := \mathbb{F}^{n\times m}$ in 3. by chasing the basis $\{u_i\}_{1\leq i\leq n}$ and $\{v_j\}_{1\leq j\leq m}$ from 1. to 3.:

- $U \otimes V$ has a basis $u_i \otimes v_j$, i.e., the $E_{i,j}$ matrices in the space $\mathbb{F}^{n\times m}$;
- An element of $U \otimes V$ is of the form $\sum \lambda_i (u_i \otimes v_i)$, i.e., the sum of $E_{i,j}$'s;
- Say a $x \in U \otimes V$ is an simple tensor, whenever there is $u \in U$ and $v \in V$ such that $x = u \otimes v$, i.e., the rank-one matrix in $\mathbb{F}^{n\times m}$;
- The tensor-rank of $x \in U \otimes V$ is the least number of simple tensors summing up to x, i.e., 使用相抵标准型将矩阵写作秩 1 矩阵的和.

Definition (The definition of tensor product in general). We consider the properties that a hypothetical object $U \otimes V$ would possess:

1. an object of $x \in U \otimes V$ is the finite sum of simple tensors, e.g., $x = \sum u_t \otimes v_t$ for some $u_t \in U$ and $v_t \in V$;
2. there is a canonical bilinear map $\theta_{U,V}: U\&V \to U \otimes V$ sending the pair (u, v) to a simple tensor $u \otimes v$, which means that

    1. $(u+u')\otimes v$ and $u\otimes v+u'\otimes v$ are the same elements,
    2. $u\otimes(v+v')$ and $u\otimes v+u\otimes v'$ are the same elements,
    3. $\lambda(u\otimes v)$, $(\lambda u)\otimes v$ and $u\otimes(\lambda v)$ are the same elements;
3. any bilinear form $U\& V\to W$ corresponds to a linear map from $U\otimes V$ to W, which means that
    1. once we have a bilinear map $B: U\& V\to W$, we have a linear map $U\otimes V\to W$ determined by the image of simple tensors:
    $$u\otimes v\mapsto B(u,v).$$
    2. once we have a linear map $\varphi: U\otimes V\to W$, there is a bilinear map $\varphi\circ\theta_{U,V}$.

> **Note**
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> In short, the linear space $\mathrm{Hom}_{\mathbb{F}}(U,\mathrm{Hom}_{\mathbb{F}}(V,W))$ (not $\mathrm{Hom}_{\mathbb{F}}(U\times V,W)$) stands for the set of bilinear maps from $U\& V$ to W. The $1:1$ correspondence means the bijection
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> $$\mathrm{Hom}_{\mathbb{F}}(U,\mathrm{Hom}_{\mathbb{F}}(V,W))\to\mathrm{Hom}_{\mathbb{F}}(U\otimes V,W),\quad B\mapsto B\circ\theta_{U,V}$$
>
> Such an bijective is also linear (verify it your self). Hence, the above is an linear isomorphism.
>
> The adjunction: $\mathrm{Hom}(A,RB)\simeq\mathrm{Hom}(LA,B)$.

Definition ($f\otimes f'$). For linear maps $f: U\to V$ and $f': U'\to V'$, the bilinear map

$$U\& U'\to V\& V'\to V\otimes V'\quad (u,u')\mapsto f(u)\otimes f'(u')$$

corresponds to a unique linear map $U\otimes U'\to V\otimes V'$ sending simple tensor $u\otimes u'$ to $f(u)\otimes f'(u')$. We denote

$$f\otimes f': U\otimes U'\to V\otimes V',\quad \sum u\otimes u'\mapsto\sum f(u)\otimes f'(u').$$

Example (The magic of $1:1$ correspondence). We show that $(g\otimes g')\circ(f\otimes f')=(g\circ f)\otimes(g'\circ f')$ are the same linear maps from $U\otimes U'$ to $W\otimes W'$, where

$$U\xrightarrow{f}V\xrightarrow{g}W,\quad U'\xrightarrow{f'}V'\xrightarrow{g'}W'.$$

> ***Step 1*** *$(g\otimes g')\circ(f\otimes f')=(g\circ f)\otimes(g'\circ f')$ whenever $(g\otimes g')\circ(f\otimes f')\circ\theta_{U,U'}=(g\circ f)\otimes(g'\circ f')\circ\theta_{U,U'}$ are the same linear map, where $\theta_{U,U'}: U\& U'\to U\otimes U'$ is the canonical bilinear map.*

> ***Step 2*** *We consider* $(g \otimes g') \circ (f \otimes f') \circ \theta_{U,U'}$. *By definition of* $f \otimes f'$, *we have* $(f \otimes f') \circ \theta_{U,U'} = \theta_{V,V'} \circ (f \& f')$, *the same bilinear map from* $U \& U'$ *to* $V \otimes V'$. *Hence*
>
> $$(g \otimes g') \circ (f \otimes f') \circ \theta_{U,U'} = (g \otimes g') \circ \theta_{V,V'} \circ (f \& f') = \theta_{W,W'} \circ (g \& g') \circ (f \& f').$$
>
> ***Step 2'*** *We consider* $(g \circ f) \otimes (g' \circ f') \circ \theta_{U,U'}$. *By definition, one has*
>
> $$(g \circ f) \otimes (g' \circ f') \circ \theta_{U,U'} = \theta_{W,W'} \circ ((g \circ f) \& (g' \circ f')).$$
>
> ***Step 3*** *It is clear that* $((g \circ f) \& (g' \circ f'))$ *and* $(g \& g') \circ (f \& f')$ *are the same thing from* $U \& U' \to W \& W'$. *This complete the proof.*

Exercise 5 Show by $1:1$ correspondence that, there exists an isomorphism

$$\Phi_{U,V,W} : U \otimes (V \otimes W) \to (U \otimes V) \otimes W.$$

(optional) Moreover, such Φ is natural, which means that for $f : U_f \to V_f$, $g : U_g \to V_g$ and $h : U_h \to V_h$, one has

$$((f \otimes g) \otimes h) \circ \Phi_{U_f,U_g,U_h} = \Phi_{V_f,V_g,V_h} \circ (f \otimes (g \otimes h))$$

Exercise 6 Show that for arbitrary linear surjection p, the linear map $\mathrm{id}_U \otimes p$ is also a surjection.

Exercise 7 Show that for arbitrary linear injection i, the linear map $\mathrm{id}_U \otimes i$ is also an injection.

> **Tip**
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> You are allowed to use the following axiom: For any vector subspace $S \subseteq L$, any linear map $f : S \to W$ extends to $\widetilde{f} : L \to W$.